Accessibility begins with access

Using AI tools to make learning materials easier to see, hear and understand

THE PROJECT

How can AI-powered captions, screen readers and text tools improve access for learners with disabilities?

PROJECT OUTPUTS

01
Tool evaluation

02
Educator guide

03
Python workflow

THE BIG IDEA

Technology should remove barriers — not create new ones.

This capstone explores practical ways AI can support accessible teaching and learning.

One campus. Different needs. Equal access.

The project focuses on learners, educators and the teams who support them.

WHO BENEFITS

01 **STUDENTS WITH DISABILITIES**

Visual, auditory and cognitive learning aids.

02 **FACULTY & EDUCATORS**

Simple workflows for accessible learning media.

03 **CAMPUS IT TEAMS**

Software compliance and infrastructure support.

PROJECT OBJECTIVES

- Benchmark 4+ free AI tools for accuracy.
- Establish WCAG 2.1 AA compliance steps.
- Automate first-draft alt-text generation.
- Keep human review in the workflow.

A toolkit for more inclusive learning

AI tools and assistive technologies work together across different formats.

01 **CHATGPT / GEMINI**

Draft image descriptions and alt-text.

02 **ELEVENLABS**

Create natural-sounding text-to-speech audio.

03 **LIVE CAPTIONS**

Transcribe lecture audio in real time.

04 **NOTEBOOKLM**

Summarize documents using W3C WAI source files.

ASSISTIVE TECH STACK

SCREEN READERS

NVDA • VoiceOver

STANDARD

W3C WCAG 2.1 AA

ACCESS MODES

Audio • Text • Captions

What the evaluation tells us

AI can speed up accessibility work, but accuracy and context still matter.

~90%	85–90%	HIGH
ALT-TEXT	TRANSCRIPTION	TEXT-TO-SPEECH
Standard images	Low-noise environments	Reliable for academic prose

LIMITATIONS & RESPONSE

01 TECHNICAL JARGON

Complex mathematical charts may contain minor interpretation errors.

02 AI HALLUCINATIONS

Low-contrast graphics may cause fabricated visual details.

03 HUMAN VERIFICATION

Check every generated description before publishing.

Project-reported estimates; actual accuracy varies by tool, content and environment.

Make accessibility part of the workflow

A repeatable checklist helps teams create and publish more inclusive materials.

THE ACCESSIBILITY CHECKLIST

- Meaningful alt-text for educational graphics.
- Synchronized captions for video lectures.
- At least 4.5:1 contrast for body text.
- Clear H1–H3 heading structure.

FROM DRAFT TO PUBLISH

01 **DRAFT** Python automation prepares alt-text prompts.

02 **REVIEW** Educator verifies accuracy and context.

03 **PUBLISH** Multi-modal materials are shared through the LMS.

Keep people at the centre

Ethics, privacy and human review are essential to responsible accessibility.

01 LIVED EXPERIENCE

Test tools with disabled users and learn from their feedback.

02 HUMAN OVERSIGHT

AI supports accessibility work; people make the final judgement.

03 PRIVACY

Protect student information when prompts and content are processed.

THE TAKEAWAY

AI can speed up captions and alt-text — but responsible accessibility needs human review.

UNIVERSAL DESIGN BENEFITS EVERY LEARNER